# Roots and Associated Structures

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# 1 Lie Algebras

## 1.1 Basic Definitions

## 1.2 Lie Algebras for the Compact Classical Lie Groups

**1.2.0.1** *$SL(n)$*

$$\mathfrak{sl}(n,\mathbb{F}) = \{X \in \mathfrak{gl}(n,\mathbb{F}) \mid \operatorname{tr} X = 0\}.$$

**1.2.0.2** *$U(n)$ & $SU(n)$*

$$\begin{aligned}
&\mathfrak{u}(n) = \{X \in \mathfrak{gl}(n,\mathbb{C}) \mid X^* = -X\},\\
&\mathfrak{t} = \{\operatorname{diag}(i\theta_1,\dots,i\theta_n) \mid \theta_i \in \mathbb{R}\},\\
&\mathfrak{g}_{\mathbb{C}} = \mathfrak{gl}(n,\mathbb{C}),\\
&\mathfrak{t}_{\mathbb{C}} = \{\operatorname{diag}(z_1,\dots,z_n) \mid z_i \in \mathbb{C}\}.
\end{aligned}$$

$$\begin{aligned}
&\mathfrak{su}(n) = \{X \in \mathfrak{gl}(\mathfrak{n},\mathbb{C}) \mid X^* = -X,\ \operatorname{tr} X = 0\},\\
&\mathfrak{t} = \{\operatorname{diag}(i\theta_1,\dots,i\theta_n) \mid \theta_i \in \mathbb{R}, \textstyle\sum_i \theta_i = 0\},\\
&\mathfrak{g}_{\mathbb{C}} = \mathfrak{sl}(n,\mathbb{C}),\\
&\mathfrak{t}_{\mathbb{C}} = \{\operatorname{diag}(z_1,\dots,z_n) \mid z_i \in \mathbb{C}, \textstyle\sum_i z_i = 0\}.
\end{aligned}$$

**1.2.0.3** *$Sp(n)$* For the first realization,

$$\mathfrak{sp}(n) = \{X \in \mathfrak{gl}(n,\mathbb{H}) \mid X^* = -X\}.$$

For the second and most used realization,

$$J = \begin{pmatrix} 0 & -I_n \\ I_n & 0 \end{pmatrix},\ \mathfrak{sp}(n,\mathbb{C}) = \{X \in \mathfrak{gl}(2n,\mathbb{C}) \mid X^t J = -JX\}.$$

$$\begin{aligned}
&\mathfrak{sp}(n) \cong \mathfrak{u}(2n) \cap \mathfrak{sp}(n,\mathbb{C}),\\
&\mathfrak{t} = \{\operatorname{diag}(i\theta_1,\dots,i\theta_n,-i\theta_1,\dots,-i\theta_n) \mid \theta_i \in \mathbb{R}\},\\
&\mathfrak{g}_{\mathbb{C}} = \mathfrak{sp}(n,\mathbb{C}),\\
&\mathfrak{t}_{\mathbb{C}} = \{\operatorname{diag}(z_1,\dots,z_n,-z_1,\dots,-z_n) \mid z_i \in \mathbb{C}\}.
\end{aligned}$$

**1.2.0.4** $SO(n)$ Define

$$T_{2m} = \frac{1}{\sqrt{2}} \begin{pmatrix} I_m & I_m \\ iI_m & -iI_m \end{pmatrix}, E_{2m} = \begin{pmatrix} 0 & I_m \\ I_m & 0 \end{pmatrix},$$

$$T_{2m+1} = \begin{pmatrix} T_{2m} & 0 \\ 0 & 1 \end{pmatrix}, E_{2m+1} = \begin{pmatrix} E_{2m} & 0 \\ 0 & 1 \end{pmatrix},$$

$$SO(E_n) = \{g \in SL(n,\mathbb{C}) \mid \bar{g} = E_n g E_n,\ g^t E_n g = E_n,\ \det g = 1\}.$$

Then

$$\mathfrak{o}(n) = \mathfrak{so}(n) = \{X \in \mathfrak{gl}(n,\mathbb{R}) \mid X^t = -X\},$$

$$\mathfrak{so}(E_n) = \{X \in \mathfrak{gl}(n,\mathbb{C}) \mid \bar{X} = E_n X E_n,\ X^t E_n + E_n X = 0\},$$

$$\mathfrak{so}(E_n)_{\mathbb{C}} = \mathfrak{so}(E_n,\mathbb{C}) = \{X \in \mathfrak{gl}(n,\mathbb{C}) \mid X^t E_n + E_n X = 0\}.$$

Working with $G = SO(E_{2n})$,

$$\mathfrak{t} = \{\mathrm{diag}(i\theta_1, \ldots, i\theta_n, -i\theta_1, \ldots, -i\theta_n) \mid \theta_i \in \mathbb{R}\},$$

$$\mathfrak{t}_{\mathbb{C}} = \{\mathrm{diag}(z_1, \ldots, z_n, -z_1, \ldots, -z_n) \mid z_i \in \mathbb{C}\}$$

Working with $G = SO(E_{2n+1})$,

$$\mathfrak{t} = \{\mathrm{diag}(i\theta_1, \ldots, i\theta_n, -i\theta_1, \ldots, -i\theta_n, 0) \mid \theta_i \in \mathbb{R}\},$$

$$\mathfrak{t}_{\mathbb{C}} = \{\mathrm{diag}(z_1, \ldots, z_n, -z_1, \ldots, -z_n, 0) \mid z_i \in \mathbb{C}\}$$

# 2 Root Theory

## 2.1 Representations of Lie Algebras

**Definition 2.1.**

- Let $\mathfrak{g}$ be the Lie algebra of a Lie subgroup of $GL(n,\mathbb{C})$. A **representation** of $\mathfrak{g}$ is a pair $(\psi, V)$, where $V$ is a finite-dimensional complex vector space and $\psi$ is a linear map $\psi : \mathfrak{g} \to \mathrm{End}(V)$, satisfying $\psi([X,Y]) = \psi(X) \circ \psi(Y) - \psi(Y) \circ \psi(X)$ for $X, Y \in \mathfrak{g}$.
- The representation $(\psi, V)$ is said to be **irreducible** if there are no proper $\psi(\mathfrak{g})$-invariant subspaces, i.e., the only $\psi(\mathfrak{g})$-invariant subspaces are 0 and $V$. Otherwise $(\psi, V)$ is called reducible.

## 2.2 Weights and Roots

## 2.3 Cartan Involution and Killing Form

**Definition 2.2.** $Z = X + iY \in \mathfrak{g}_{\mathbb{C}}$, $X, Y \in \mathfrak{g}$. The **Cartan involution** $\theta$ is given by $\theta(Z) = X - iY$.

Check that $\theta Z = -Z^*$.

**Definition 2.3.** For $X, Y \in \mathfrak{g}_{\mathbb{C}}$, the symmetric complex bilinear form $B(X,Y) = \mathrm{tr}(\mathrm{ad}X \circ \mathrm{ad}Y)$ on $\mathfrak{g}_{\mathbb{C}}$ is called the **Killing form**.

## 2.4 The Standard $\mathfrak{sl}(2,\mathbb{C})$ and $\mathfrak{su}(2)$ Triples

Let $G$ be a compact Lie algebra and $\mathfrak{t}$ a Cartan subalgebra of $\mathfrak{g}$. When $\mathfrak{g}$ is semisimple, recall that $B$ is negative definite on $\mathfrak{t}$. It follows that $B$ restricts to a real inner product on the real vector space $i\mathfrak{t}$. Continuing to write $(i\mathfrak{t})^*$ for the set of $\mathbb{R}$-linear functionals on $i\mathfrak{t}$, $B$ induces an isomorphism between $i\mathfrak{t}$ and $(i\mathfrak{t})^*$ as follows.

**Definition 2.4.** Let $G$ be a compact Lie group with semisimple Lie algebra. Let $u_\alpha \in i\mathfrak{t}$ be uniquely determined by the equation

$$\alpha(H) = B(H, u_\alpha)$$

for all $H \in i\mathfrak{t}$ and, when $\alpha \neq 0$, let

$$h_\alpha = \frac{2u_\alpha}{B(u_\alpha, u_\alpha)}.$$

In case $\mathfrak{g}$ is not semisimple, define $u_\alpha \in i\mathfrak{t}' \in i\mathfrak{t}$. Note that the equation $\alpha(H) = B(H, u_\alpha)$ now holds for all $H \in \mathfrak{t}_{\mathbb{C}}$ by $\mathbb{C}$-linear extension. An alternate notation for $h_\alpha$ is $\alpha^\vee$.

Check that $h_\alpha$ can be computed with respect to the trace form instead of the Killing form.

**Lemma 2.5.** *Let $G$ be a compact Lie group, $\mathfrak{t}$ a Cartan subalgebra of $\mathfrak{g}$, and $\alpha \in \Delta(\mathfrak{g}_{\mathbb{C}})$.*

- *For $E \in \mathfrak{g}_\alpha$ and $F \in \mathfrak{g}_{-\alpha}$,*

$$[E,F] = B(E,F)u_\alpha = \frac{1}{2}B(E,F)B(u_\alpha,u_\alpha)h_\alpha.$$

- *Given a nonzero $E \in \mathfrak{g}_\alpha$, $E$ may be rescaled by an element of $\mathbb{R}$, so that $[E, -\theta E] = h_\alpha$.*

For the next theorem, recall that $B$ is nonsingular on $\mathfrak{g}_\alpha \times \mathfrak{g}_{-\alpha}$ and that $-B(X, \theta Y)$ is an Ad-invariant inner product on $\mathfrak{g}'_{\mathbb{C}}$ for $X, Y \in \mathfrak{g}_{\mathbb{C}}$.

**Theorem 2.6.** *Let $G$ be a compact Lie group, $\mathfrak{t}$ a Cartan subalgebra of $\mathfrak{g}$, and $\alpha \in \Delta(\mathfrak{g}_{\mathbb{C}})$. Fix a non zero $E_\alpha \in \mathfrak{g}_\alpha$ and let $F_\alpha = -\theta E_\alpha$. Rescale $E_\alpha$ (and therefore $F_\alpha$), so that $[E_\alpha, F_\alpha] = H_\alpha$ where $H_\alpha = h_\alpha$.*

- *$\mathfrak{sl}(2,\mathbb{C}) \cong span_{\mathbb{C}}(E_\alpha, H_\alpha, F_\alpha)$ with $\{E_\alpha, H_\alpha, F_\alpha\}$ corresponding to the standard basis*

$$E = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix},\ H = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix},\ F = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$$

*of $\mathfrak{sl}(2,\mathbb{C})$.*

- *Let* $\mathcal{I}_\alpha = iH_\alpha$, $\mathcal{J}_\alpha = -E_\alpha + F_\alpha$, $\mathcal{K}_\alpha = -i(E_\alpha + F_\alpha)$. *Then* $\mathcal{I}_\alpha, \mathcal{J}_\alpha, \mathcal{K}_\alpha \in \mathfrak{g}$ *and* $\mathfrak{su}(2) \cong \operatorname{span}_{\mathbb{R}}\{\mathcal{I}_\alpha, \mathcal{J}_\alpha, \mathcal{K}_\alpha\}$ *with* $\{\mathcal{I}_\alpha, \mathcal{J}_\alpha, \mathcal{K}_\alpha\}$ *corresponding to the basis*

$$\begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}, \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & -i \\ -i & 0 \end{pmatrix}$$

  *of* $\mathfrak{su}(2)$.

- *There exists a Lie algebra homomorphism* $\varphi_\alpha : SU(2) \to G$, *so that* $d\varphi : \mathfrak{su}(2) \to \mathfrak{g}$ *implements the iso in the 2nd and whose complexification* $d\varphi : \mathfrak{sl}(2, \mathbb{C}) \to \mathfrak{g}_{\mathbb{C}}$ *implements the iso in the 1st.*

- *The image of* $\varphi_\alpha$ *in* $G$ *is a Lie subgroup of* $G$ *isomorphic to either* $SU(2)$ *or* $SO(3)$ *depending on whether the kernel of* $\varphi_\alpha$ *is* $\{I\}$ *or* $\{\pm I\}$.

(Recall: complexificate $\mathfrak{su}(2)$ getting $\mathfrak{sl}(2, \mathbb{C})$, then 'diagonalize' $H$ getting $\{E, H, F\}$.)